

LIGHT CONTROLLED LED CIRCUIT

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INTRODUCTION

This project presents a simple **light-controlled lighting system** using a photoresistor (LDR) and a transistor.

The aim is to design a circuit that intelligently responds to changes in ambient light conditions, demonstrating basic principles of sensing and switching.



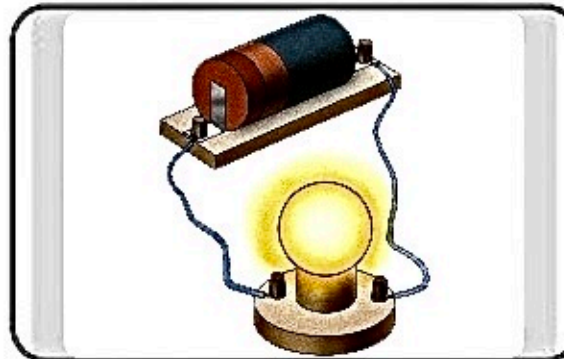
POWER & CONNECTIVITY

Adding Batteries in Series - Brightness of Bulbs



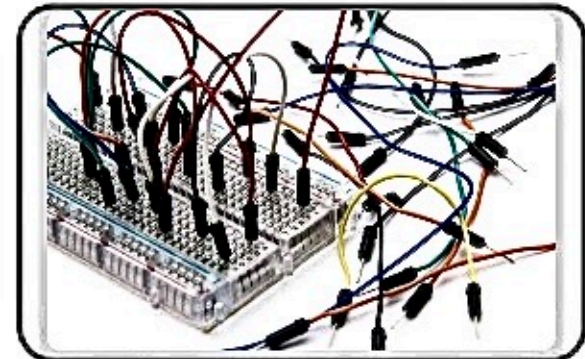
9V Battery

Supplies power to the entire circuit.



Battery Holder

Provides a stable and secure connection.



Jumper Wires

Used for flexible connections on breadboard.

CORE COMPONENTS

2N2222 Transistor

A small NPN switching transistor used to amplify and control current flow. It acts as the "brain" that turns the LED on or off.

Medium Breadboard

The main platform to assemble and organize components, making it easy to test and modify the circuit without soldering.

SENSING ELEMENTS



LDR

Changes resistance based on light levels. High in dark, low in light.



22K Resistor

Stabilizes current flow and balances the output signal.



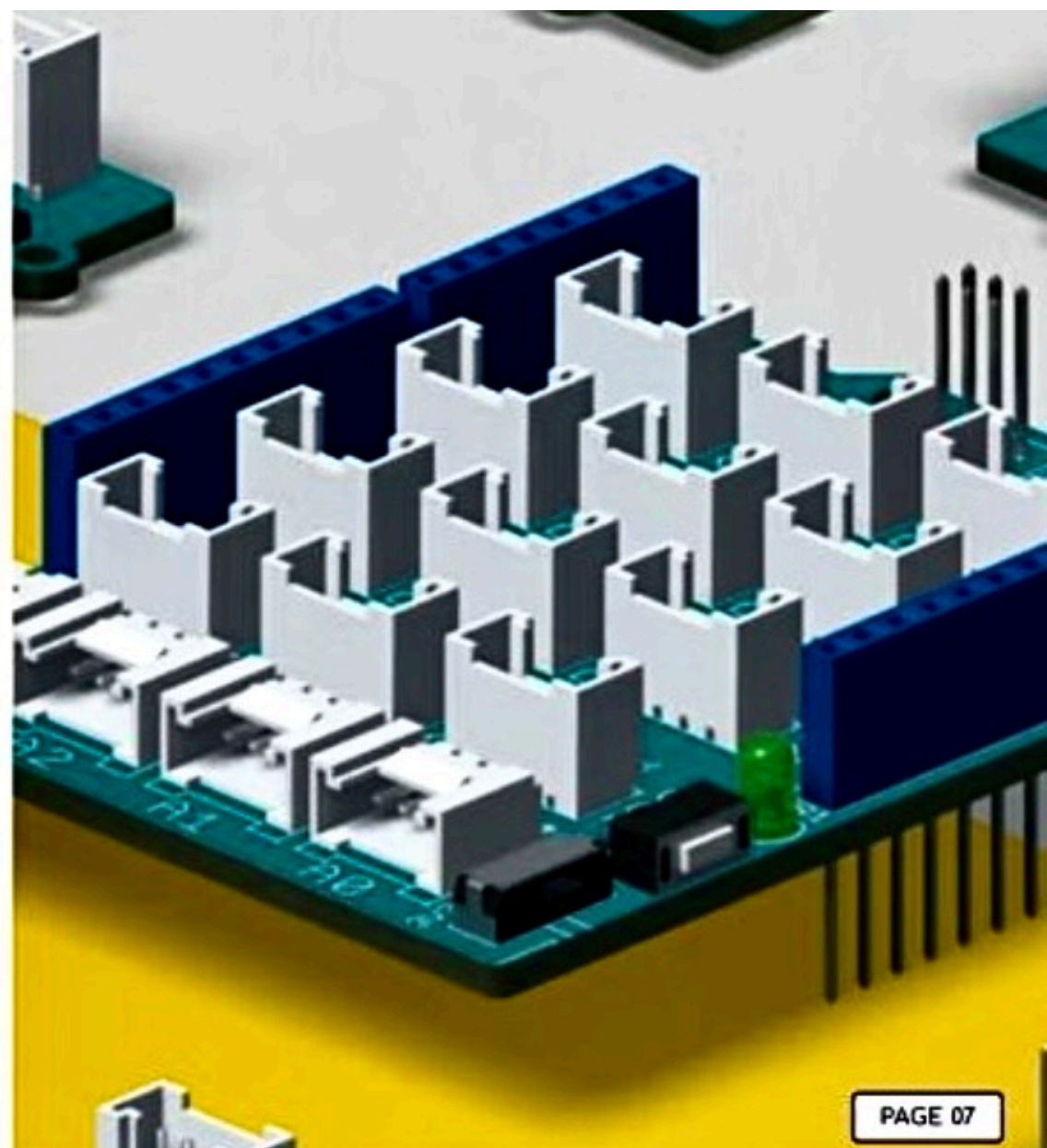
LED

The output device that converts electrical energy into light.

THE WORKING PRINCIPLE

SCENARIO A: DARKNESS

- LDR senses darkness.
- Resistance increases significantly.
- Voltage reaches the transistor base.
- **Transistor turns ON.**
- **LED LIGHTS UP.**



SCENARIO B: BRIGHT LIGHT

In bright light, the **LDR resistance decreases**, which reduces the voltage flowing to the transistor base.

Result: The transistor turns **OFF**, and the LED turns off automatically.



DAILY LIFE APPLICATIONS

01

Smart Street Lights: Automating city lights to save energy.

02

Garden Lighting: Lamps that turn on at sunset automatically.

03

Security Systems: Sensors that detect intruders by light obstruction.



RESULTS & CONCLUSION

The project successfully demonstrated an automated lighting system that responds to environmental changes. We conclude that:

- The LDR effectively acts as a sensitive input controller.
- The Transistor serves as an efficient electronic switch.
- The system achieves **Energy Efficiency** by operating only when needed (in the dark).

THE RESEARCH TEAM



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THANK YOU

WE HOPE THIS EXPLANATION CLARIFIED THE
CONCEPT OF LIGHT-CONTROLLED CIRCUITS.



IMAGE SOURCES



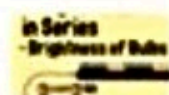
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Source: stockcake.com



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Source: store-usa.arduino.cc



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Source: www.youtube.com



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Source: sensorkit.arduino.cc



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